



# SOCIAL MEDIA ANALYTICS DASHBOARD

## Graduation Project Documentation

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## ACKNOWLEDGMENT

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Their valuable feedback and encouragement helped us successfully complete this work.

We also thank all team members for their dedication, collaboration, and hard work.

## Introduction

Social media platforms generate massive amounts of data every day. Organizations need effective tools to analyse this information and transform it into valuable business insights.

This project presents a comprehensive Social Media Analytics Dashboard developed using Microsoft Power BI to monitor audience engagement, campaign performance, content effectiveness, and business growth indicators.

## Project Overview

The Social Media Analytics Dashboard is designed to provide businesses and marketing teams with a centralized platform for monitoring social media activities across multiple channels.

The dashboard enables users to:

- Track engagement performance.
- Measure audience interactions.
- Analyse viral trends.
- Evaluate campaign effectiveness.
- Support strategic decision-making through data analytics.

## Project Overview

Organizations face significant challenges in managing and analyzing large volumes of social media data. Manual analysis is often time-consuming, inefficient, and prone to errors.

The lack of centralized reporting makes it difficult to identify audience behaviour patterns, campaign performance, and emerging trends.

This project addresses these challenges by providing an interactive analytical dashboard that transforms raw social media data into actionable insights.

## Project Objectives

The primary objectives of the project are:

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- ❖ Monitor views, likes, comments, and shares.
- ❖ Measure audience engagement levels.
- ❖ Identify viral content and trending hashtags.
- ❖ Analyse campaign performance.
- ❖ Understand audience behaviour.
- ❖ Support data-driven business decisions.
- ❖ Improve marketing effectiveness.

## Tools and Technologies

The project was developed using the following technologies:

- **Microsoft Power BI**  
Used for data visualization and dashboard development.
- **DAX (Data Analysis Expressions)**  
Used for creating calculated measures and KPIs.
- **Data Modelling**  
Used to establish relationships between tables.
- **Data Cleaning**  
Used to improve data quality and consistency.
- **Interactive Dashboards**

Used to provide dynamic reporting and filtering capabilities.

## Data Preparation Process

The data preparation process consisted of several stages:

### ➤ Data Collection

Gathering social media datasets from multiple sources.

### ➤ Data Cleaning

- Removing duplicates.
- Handling missing values.
- Standardizing language and country fields.
- Correcting inconsistent labels.

### ➤ **Data Processing**

Transforming raw data into structured datasets.

### ➤ **Data Modelling**

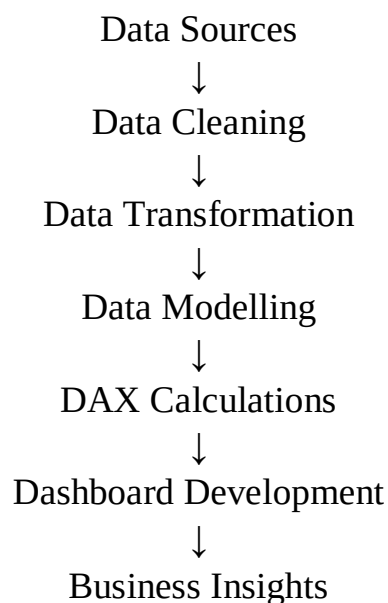
Building relationships between fact and dimension tables.

### ➤ **Data Visualization**

Creating dashboards and business reports.

## **System Architecture**

The system architecture follows the Business Intelligence workflow:



This architecture ensures accurate reporting and reliable decision-making.

## **Dashboard Overview**

The dashboard contains multiple analytical sections:

### ➤ **Performance Dashboard**

Tracks views, likes, and interactions.

### ➤ **Engagement Dashboard**

Measures audience engagement rates.

### ➤ **Campaign Dashboard**

Evaluates marketing campaign effectiveness.

## ➤ Viral Trends Dashboard

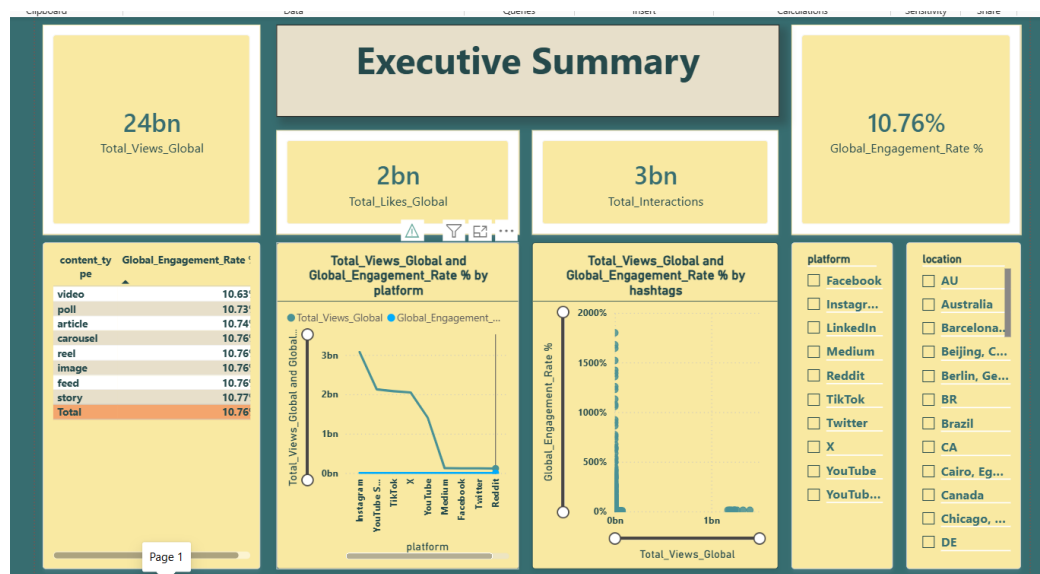
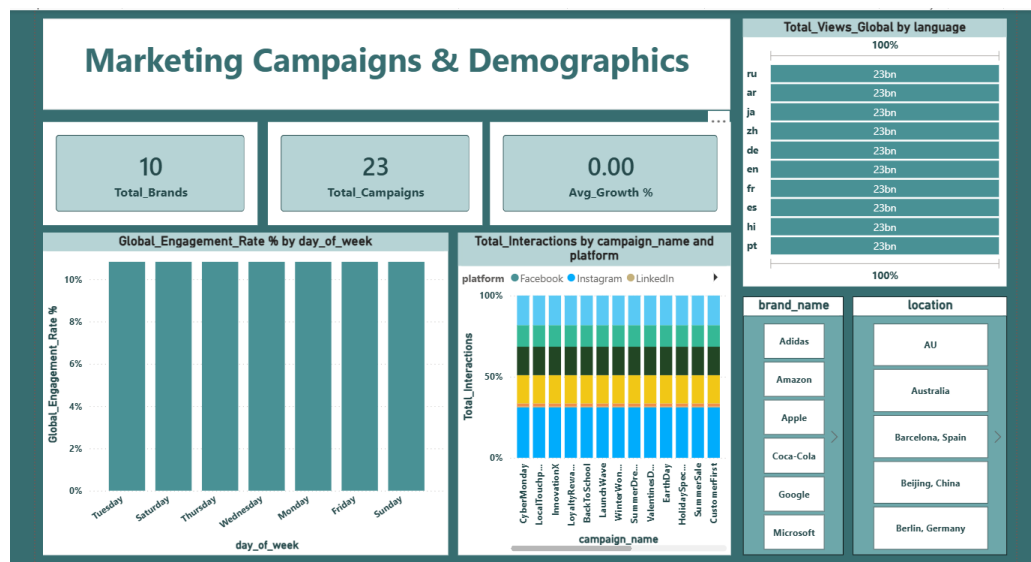
Analyses trending content and hashtags.

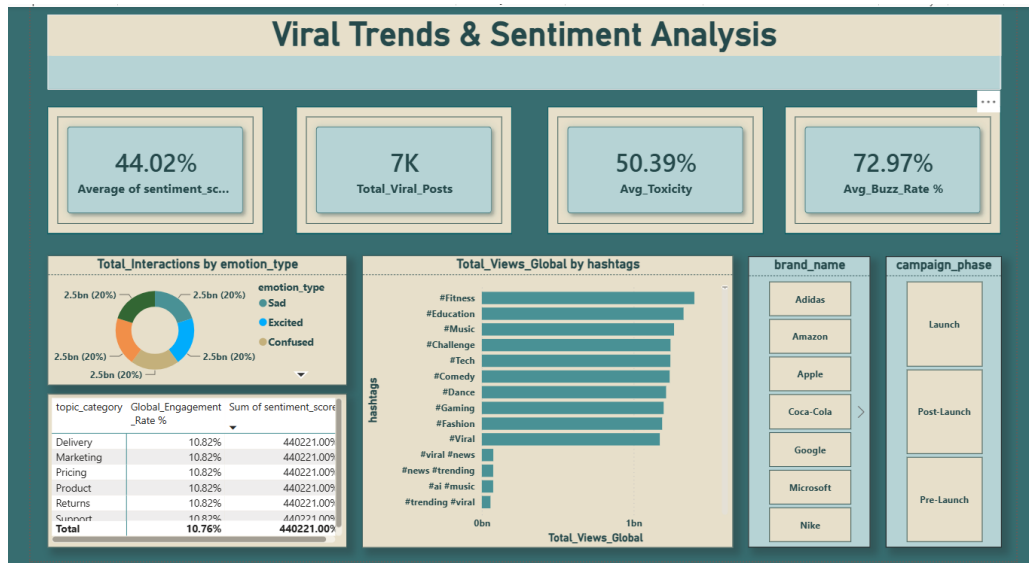
## ➤ Sentiment Dashboard

Measures audience sentiment and toxicity levels.

## ➤ Demographics Dashboard

Provides insights about audience language and geographic distribution.





## Key Performance Indicators (KPIs)

The dashboard includes several important KPIs:

- **Total Views Global**  
Measures total content visibility.
- **Total Likes Global**  
Measures audience appreciation and reactions.
- **Total Interactions**  
Calculates likes, comments, and shares.
- **Global Engagement Rate**  
Measures audience engagement efficiency.
- **Average Buzz Rate**  
Measures audience interest and trending discussions.
- **Average Toxicity**  
Measures negative sentiment levels.
- **Total Viral Posts**  
Counts high-performing content.
- **Total Campaigns**  
Measures the number of campaigns analyzed.
- **Total Brands**  
Measures the number of brands included.
- **Average Growth Rate**  
Measures audience and campaign growth.



## Business Insights

### ➤ Platform Performance Analysis

Instagram and YouTube Shorts generate the highest visibility and audience reach.

LinkedIn and X (Twitter) achieve stronger engagement rates despite lower view counts.

### ➤ Content Format Analysis

Video and Carousel content outperform static content by generating significantly higher engagement rates.

### ➤ Sentiment Analysis

Positive emotions drive higher shares and content virality.

High Buzz Rates are often associated with increased toxicity levels, requiring monitoring and moderation.

### ➤ Campaign Performance Analysis

Mid-week publishing schedules generate higher engagement rates.

Post-launch campaign activities significantly increase audience retention.

### ➤ Demographic Insights

Arabic and English content achieve the highest engagement levels, supporting regional expansion opportunities.

## Challenges and Solutions

### ➤ Challenge 1

Data inconsistency across platforms.

#### **Solution**

Applied data cleaning and standardization techniques.

### ➤ Challenge 2

Missing and duplicated records.

### **Solution**

Implemented data validation and cleansing procedures.

### ➤ **Challenge 3**

Weak relationships between tables.

### **Solution**

Redesigned the data model using dimension and fact tables.

### ➤ **Challenge 4**

Large volume of social media data.

### **Solution**

Utilized Power BI optimization techniques and DAX calculation

## **Future Enhancements**

Future development opportunities include:

- Real-time social media monitoring.
- AI-powered sentiment analysis.
- Predictive analytics and forecasting.
- Integration with additional platforms.
- Mobile dashboard support.
- Automated reporting and alerts.

## **Conclusion**

The Social Media Analytics Dashboard successfully demonstrates the power of data analytics in understanding audience behaviour and evaluating marketing performance.

The project provides organizations with valuable insights that improve decision-making, increase engagement, optimize marketing investments, and support long-term business growth.

The dashboard serves as a strong foundation for future business intelligence and advanced analytics initiatives